

HW 6

March 30, 2023

Problem 0.1 (Chapter 13, problem 2a). Let p be a prime number and k a positive integer. Show that if x is an integer such that $x^2 \equiv x \pmod{p}$, then $x \equiv 0$ or $1 \pmod{p}$.

Proof. Suppose that $x^2 \equiv x \pmod{p}$. Suppose that x is not congruent to 0 modulo p . Then there exists a y such that $xy \equiv 1 \pmod{p}$. Multiplying both sides by y , we get $x \equiv 1 \pmod{p}$. We conclude that either $x \equiv 0$ or $x \equiv 1 \pmod{p}$. $\square$

Problem 0.2 (Chapter 13, problem 8). Find $\bar{x}, \bar{y} \in \mathbb{Z}_{15}$ such that $\bar{x}\bar{y} = \bar{0}$ but $\bar{x} \neq \bar{0}$ and $\bar{y} \neq \bar{0}$. Find a condition on m such that the equality $\bar{x}\bar{y} = \bar{0}$ in $\mathbb{Z}_m$ implies that either $\bar{x} = \bar{0}$ or $\bar{y} = \bar{0}$.

Proof. $\bar{x} = \bar{3}$ and $\bar{y} = \bar{5}$ have product $\bar{3}\bar{5} = \bar{0}$.

Suppose that m is prime. We will show that if $\bar{x}\bar{y} = \bar{0}$ in $\mathbb{Z}_m$ then either $\bar{x} = \bar{0}$ or $\bar{y} = \bar{0}$. Indeed, suppose that x and y are integers such that $0 \leq x, y \leq m-1$. If $\bar{x}\bar{y} = \bar{0}$, then $xy \equiv 0 \pmod{m}$, so $xy = mk$ for some integer k . Thus, m divides xy . As m is prime, it must be the case that m divides either x or y . Therefore at least one of x and y is 0. $\square$

Problem 0.3 (Chapter 14, problem 2). Let a, b be integers, and let p be a prime not dividing a . Show that the solution of the congruence equation $ax \equiv b \pmod{p}$ is $x \equiv a^{p-2}b \pmod{p}$. Use this observation to solve the congruence equation $4x \equiv 11 \pmod{19}$.

Proof. By Fermat's little theorem, we know that $a^{p-1} \equiv 1 \pmod{p}$. This implies that

$$a(a^{p-2}b) \equiv a^{p-1}b \equiv b \pmod{p}$$

Now let's solve $4x \equiv 11 \pmod{19}$. By the above result, we have $x \equiv 4^{17} \cdot 11 \pmod{19}$. We can compute this using repeated squaring:

$$4^2 \equiv 16 \pmod{19}$$

$$4^4 \equiv 9 \pmod{19}$$

$$4^8 \equiv 5 \pmod{19}$$

$$4^{16} \equiv 6 \pmod{19}$$

We now compute

$$4^{17} \cdot 11 \equiv 4^{16} \cdot 4 \cdot 11 \equiv 6 \cdot 4 \cdot 11 \equiv 17 \pmod{19}$$

So the answer is $x \equiv 17$. $\square$